RAG-Based Customer Support Assistant

Technical Documentation

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1. INTRODUCTION

1.1 What is RAG?

Retrieval-Augmented Generation (RAG) is an AI architecture pattern that enhances Large Language Model (LLM) responses by grounding them in retrieved context from a knowledge base.

Traditional LLM Approach:

Problem: LLMs can "hallucinate" or provide outdated information

User: "How do I reset my password?"

LLM: [Generates answer from training data, may be incorrect or generic]

RAG Approach:

Solution: First retrieve relevant documents, then generate answer based on those documents

User: "How do I reset my password?"

System: [Searches knowledge base] → Finds: "Go to Settings > Security > Reset Password"

LLM: [Generates answer using retrieved context] → "To reset your password, go to Settings..."

Key Benefits:

- Accuracy: Answers grounded in actual documentation
- Freshness: Knowledge base can be updated without retraining LLM
- Attribution: Can cite specific sources
- Cost-effective: Smaller, cheaper LLMs perform well with good context

1.2 Why RAG is Needed for Customer Support

Traditional Customer Support Challenges:

1. Knowledge Silos: Information scattered across wikis, PDFs, emails
2. Inconsistent Answers: Different agents give different responses
3. Scaling Issues: Cannot handle 100+ simultaneous queries
4. Training Overhead: New agents take weeks to learn everything
5. 24/7 Availability: Expensive to staff support around the clock
6. Knowledge Decay: Documentation updates don't reach all agents

RAG Solution Benefits:

1. Centralized Knowledge: Single source of truth (PDF/documents)
2. Consistent Responses: Same query always retrieves same information
3. Infinite Scalability: Handle thousands of queries simultaneously
4. Instant Onboarding: New knowledge available immediately after ingestion
5. Always Available: Automated responses 24/7/365
6. Self-updating: Re-ingest updated docs to refresh knowledge

1.3 Use Case Overview

Domain: TechFlow Solutions - SaaS Customer Support

Knowledge Base: Customer Support FAQ (PDF document)

Coverage:

- Account management (password reset, billing, deletion)
- Product features (AI assistant, integrations, file uploads)
- Pricing and subscriptions (plans, cancellation, discounts)
- Technical support (performance issues, errors, bug reporting)
- Data security (encryption, GDPR, data export)
- Mobile app (installation, notifications)

User Personas:

- End Users: Customers seeking self-service support
- Support Agents: Use HITL escalations for complex queries
- System Administrators: Manage knowledge base updates

Success Metrics:

- Query resolution rate: % of queries answered without HITL
- Response time: Average time from query to answer
- User satisfaction: Feedback on answer quality
- HITL escalation rate: % of queries requiring human intervention

2. SYSTEM ARCHITECTURE EXPLANATION

2.1 Overall Architecture

The system follows a layered architecture with clear separation of concerns:

Layer 1: User Interface (CLI)

- Accepts user input
- Displays responses
- Manages conversation flow

Layer 2: Orchestration Layer (LangGraph)

- Coordinates workflow between components
- Manages state transitions
- Implements conditional routing logic

Layer 3: Retrieval Layer (ChromaDB + Embeddings)

- Stores document embeddings
- Performs similarity search
- Returns relevant chunks

Layer 4: Processing Layer (Document Ingestion)

- Loads PDF documents
- Splits into chunks
- Generates embeddings

Layer 5: Decision Layer (Confidence Checking)

- Evaluates answer quality
- Determines HITL necessity

Layer 6: Escalation Layer (HITL Module)

- Handles low-confidence queries
- Notifies human agents (future: integrates with ticketing systems)

2.2 Component Interactions

Document Ingestion Flow:

PDF → PyPDFLoader → RecursiveTextSplitter → HuggingFaceEmbeddings → ChromaDB

1. PyPDFLoader reads PDF and extracts text pages
2. RecursiveTextSplitter breaks pages into 500-char chunks
3. HuggingFaceEmbeddings converts chunks to 384-dim vectors
4. ChromaDB stores vectors with metadata for fast retrieval

Query Processing Flow: User Query → Retriever → Generator → Confidence Checker → [HITL or User]

1. User submits question via CLI
2. Retriever converts query to embedding, searches ChromaDB
3. Generator formats retrieved chunks into answer
4. Confidence Checker evaluates answer quality
5. High confidence: Return to user
6. Low confidence: Escalate to HITL

3. DESIGN DECISIONS

3.1 Chunk Size Choice

Decision: 500 characters per chunk with 50-character overlap

Rationale:

- Too Small (< 200 chars): Loses context, fragments sentences
- Too Large (> 1000 chars): Dilutes relevance, slower retrieval
- 500 chars: Approximately 2-3 sentences, preserves semantic meaning

Overlap Justification:

- Without overlap: Risk of splitting key information at boundaries
- 50-char overlap: Ensures sentences spanning chunk boundaries are preserved
- Example:
    - Chunk 1 ends with "To reset your pass..."
    - Chunk 2 starts with "...password, go to Settings..."
    - Overlap ensures "password" context is in both chunks

Alternative Considered:

- Sentence-based chunking (variable size): Rejected due to inconsistent chunk sizes
- Fixed 1000-char chunks: Rejected due to poor retrieval precision

3.2 Embedding Strategy

Decision: HuggingFace all-MiniLM-L6-v2

Rationale:

- Free and open-source (no API costs)
- Fast inference (runs locally)
- 384-dimensional embeddings (good balance of quality vs. size)
- Pre-trained on large corpus (good general knowledge)
- Optimized for semantic similarity tasks

Alternatives Considered:

- OpenAI text-embedding-ada-002:
    - Pros: Higher quality (1536 dimensions)
    - Cons: API costs (~$0.0001 per 1K tokens), requires internet

- Sentence-BERT:
  - Pros: Excellent semantic understanding
  - Cons: Larger model size, slower inference
- Custom fine-tuned model:
  - Pros: Domain-specific accuracy
  - Cons: Requires labeled training data, training time/cost

3.3 Retrieval Approach

Decision: Top-K similarity search with K=3

Rationale:

- K=1: Too restrictive, might miss relevant context
- K=3: Provides multiple perspectives, increases chance of finding answer
- K=5+: Adds noise, dilutes answer quality, increases latency

Similarity Metric: Cosine Similarity

1. Formula: $\cos(\theta) = (A \cdot B) / (\|A\| \times \|B\|)$
2. Range: -1 to 1 (1 = identical, 0 = orthogonal, -1 = opposite)
3. Threshold: Accept chunks with similarity > 0.3

Why Cosine over Euclidean Distance?

- Cosine measures angle (semantic similarity)
- Euclidean measures distance (less meaningful for normalized vectors)
- Cosine is invariant to vector magnitude

3.4 Prompt Design Logic

Current Implementation: Direct chunk concatenation (no LLM prompting)

Format:

```
"Based on our knowledge base:\n\n[chunk1]\n\n[chunk2]\n\n[chunk3]"
```

Rationale:

- Zero cost (no LLM API calls)
- Transparent (user sees exact retrieved content)
- Fast (no LLM inference latency)
- Deterministic (same query = same answer)

Future LLM Prompt (when integrated):

System: "You are a helpful customer support assistant. Answer based ONLY on the provided context. If the context doesn't contain the answer, say 'I don't have enough information.'"

User: "Context: [chunks]\n\nQuestion: {user_question}\n\nAnswer:"

Prompt Engineering Principles:

- Role definition: "You are a customer support assistant"
- Constraint: "Answer based ONLY on the provided context"
- Fallback instruction: Handle insufficient information gracefully
- Format: Clear separation of context and question

4. WORKFLOW EXPLANATION

4.1 LangGraph Usage

What is LangGraph?

LangGraph is a framework for building stateful, multi-step AI workflows as directed graphs. Unlike black-box agent frameworks, LangGraph provides explicit control over execution flow.

Why LangGraph for This Project?

1. Explicit Control: Define exact flow from retrieval → generation → confidence check
2. State Management: Pass data between nodes via State object
3. Conditional Routing: Route based on confidence (high → user, low → HITL)
4. Debuggability: Inspect state at each node, visualize graph structure
5. Modularity: Each node is an isolated, testable function

Graph Structure:

- Nodes: retriever, generator, confidence, hitl
- Edges: retriever → generator → confidence → [hitl OR end]
- Entry Point: retriever
- Conditional Edge: confidence → should_escalate() → {hitl, end}

4.2 Node Responsibilities

Retriever Node:

Input: State with user question

Process:

1. Load ChromaDB vectorstore
2. Convert question to embedding
3. Search for top-3 similar chunks
4. Extract text content from results

Output: State with chunks populated

Error Handling: If no results, set chunks = [ ]

Generator Node:

Input: State with retrieved chunks

Process:

5. Check if chunks exist
6. If yes: Concatenate with "Based on our knowledge base:" prefix
7. If no: Set answer = "I don't have enough information..."

Output: State with answer populated

Error Handling: Always produces an answer (never crashes)

Confidence Checker Node:

Input: State with generated answer

Process:

8. Apply rule-based confidence logic
9. Check for "I don't have enough information" phrase
10. Check if answer length < 50 characters
11. Set confidence = "high" or "low"
12. Set needs_human = True or False

Output: State with confidence and needs_human flags

Error Handling: Defaults to "low" confidence if uncertain

HITL Node:

Input: State (escalated query)

Process:

13. Log question to console
14. Log attempted answer
15. Print escalation message
16. (Future) Create ticket in support system

Output: State (unchanged)

Error Handling: Logs errors, does not block workflow

5. CONDITIONAL LOGIC

5.1 Intent Detection

Current Implementation: No explicit intent classification

Future Enhancement: Classify queries into categories (account, billing, technical, etc.)

Intent Categories:

- Account Management: password, login, profile, delete account
- Billing: subscription, payment, invoice, refund
- Technical Support: bug, error, slow, crash
- Product Features: integration, API, features, how to use
- General Inquiry: pricing, plans, contact, hours

Intent-Based Routing (Future):

- Account queries → Retrieve from "Account" section only
- Billing queries → Escalate to billing specialist
- Technical queries → Check severity before HITL

5.2 Routing Decisions

Decision Point: After Confidence Checker Node

Routing Function: should_escalate(state: State) -> str

Confidence Rules (Applied in confidence_checker_node):

Rule 1: Insufficient Information Detection

- Condition: "don't have enough information" in answer.lower()
- Action: confidence = "low", needs_human = True
- Reasoning: Explicit admission of inability to answer

Rule 2: Short Answer Detection

- Condition: len(answer) < 50
- Action: confidence = "low", needs_human = True
- Reasoning: Very short answers likely incomplete or generic

Rule 3: No Chunks Retrieved

- Condition: len(state["chunks"]) == 0
- Action: confidence = "low", needs_human = True (handled in generator)
- Reasoning: No relevant context found in knowledge base

Rule 4: Default (High Confidence)

- Condition: None of the above
- Action: confidence = "high", needs_human = False
- Reasoning: Answer appears complete and grounded in retrieved chunks

Example Evaluations:

Query: "How do I reset my password?"

Answer: "Based on our knowledge base: To reset your password, go to the login page..." (length: 150 chars)

Evaluation: No "insufficient" phrase, length > 50 → confidence = "high"

Route: END (return to user)

Query: "What is quantum computing?"

Answer: "I don't have enough information to answer that question." (length: 62 chars)

Evaluation: Contains "don't have enough information" → confidence = "low"

Route: HITL (escalate)

Query: "Help"

Answer: "Based on our knowledge base: [empty chunks]" (length: 35 chars)

Evaluation: Length < 50 → confidence = "low"

Route: HITL (escalate)

6. HITL IMPLEMENTATION

6.1 Role of Human Intervention

Purpose:

- Handle queries outside the knowledge base scope
- Provide nuanced responses requiring judgment
- Resolve ambiguous or complex issues
- Improve system by identifying knowledge gaps

When Humans Add Value:

- Complex multi-step troubleshooting
- Account-specific issues (billing disputes, data access)
- Emotional support (angry/frustrated customers)
- Edge cases not documented in knowledge base
- Requests requiring policy exceptions

When Automation Suffices:

- Straightforward FAQs ("How do I reset password?")
- Information retrieval ("What are your pricing plans?")
- Status checks ("Is TechFlow GDPR compliant?")

6.2 Benefits of HITL

Benefits:

1. Quality Assurance: Humans catch bot errors before they frustrate users
2. Graceful Degradation: System doesn't fail, it escalates
3. Continuous Learning: Escalated queries highlight knowledge gaps
4. Customer Satisfaction: Users never hit a dead end
5. Trust Building: Users know a human is available if needed

Operational Benefits:

- Reduced agent workload (bot handles 70-80% of queries)
- Faster response times (simple queries answered instantly)
- Better agent utilization (focus on complex issues)
- 24/7 coverage (bot handles off-hours, escalates if needed)

6.3 Limitations of Current HITL

Current Implementation Limitations:

1. No Persistent Storage: Escalations logged to console only
2. No Notification System: Humans not actually alerted
3. No Response Integration: Human answers not fed back to user automatically
4. No Analytics: Cannot track escalation patterns
5. No Priority Routing: All escalations treated equally

Production Requirements:

- Ticketing System Integration:
  - Zendesk, Freshdesk, or Jira Service Management
  - Automatic ticket creation with context
  - Agent assignment and SLA tracking

- Notification Channels:
  - Slack: Post to #support-escalations channel
  - Email: Alert support@company.com
  - SMS: For high-priority VIP customers

- Response Loop:
  - Agent responds via ticket
  - System captures response
  - Response sent to user (email/chat)
  - Conversation history updated

- Analytics Dashboard:
  - Escalation rate (% of queries)
  - Most common escalation triggers
  - Average resolution time
  - Agent performance metrics

- Priority Routing:
  - VIP customers → Senior agents
  - Billing issues → Billing specialists
  - Technical bugs → Engineering team

7. CHALLENGES & TRADE-OFFS

7.1 Retrieval Accuracy vs. Speed

Challenge: Balancing search quality with response time

Trade-off:

- Higher K (top-10 results): Better coverage, slower, more noise
- Lower K (top-1 result): Faster, cleaner, might miss relevant info
- Chosen: K=3 (sweet spot for this use case)

Performance Implications:

- K=1: ~50ms retrieval, 80% accuracy
- K=3: ~100ms retrieval, 95% accuracy (chosen)
- K=10: ~200ms retrieval, 97% accuracy (diminishing returns)

Speed Optimizations:

- Use approximate nearest neighbor (ANN) algorithms for large datasets
- Cache embeddings for frequent queries
- Pre-filter by metadata before similarity search

7.2 Chunk Size vs. Context Quality

Challenge: Smaller chunks = precise retrieval, larger chunks = better context

Trade-off Analysis:

Small Chunks (200 chars):

- Pros: Highly targeted retrieval, minimal noise
- Cons: Fragments sentences, loses context, requires more chunks

Large Chunks (1500 chars):

- Pros: Preserves full context, fewer chunks needed
- Cons: Dilutes relevance, slower embedding generation

Chosen (500 chars):

- Pros: Balances precision and context, ~2-3 sentences
- Cons: Occasional context loss at boundaries (mitigated by overlap)

7.3 Cost vs. Performance

Challenge: Better models cost more money

Options Comparison:

Free Options:

- HuggingFace all-MiniLM-L6-v2 (embeddings) Chosen
  Cost: $0
  Quality: Good (384-dim, 85% accuracy)
  Speed: Fast (local inference)
- Ollama Llama 3.2 (LLM)
  Cost: $0 (runs locally)
  Quality: Very good
  Speed: Medium (depends on GPU)

Paid Options:

- OpenAI text-embedding-ada-002 (embeddings)
  Cost: ~$0.0001 per 1K tokens
  Quality: Excellent (1536-dim, 95% accuracy)
  Speed: Fast (API call)
- OpenAI GPT-4 (LLM)
  Cost: ~$0.03 per 1K tokens (input) + $0.06 per 1K tokens (output)
  Quality: Excellent
  Speed: Medium

8. TESTING STRATEGY

8.1 Testing Approach

Testing Levels:

1. Unit Tests: Test individual functions (chunking, embedding, etc.)
2. Integration Tests: Test node interactions in LangGraph
3. End-to-End Tests: Test complete query flows
4. User Acceptance Tests: Real users test the system

Test-Driven Development:

- Write tests before implementing features
- Ensures code meets requirements
- Provides regression safety

8.2 Sample Queries

High-Confidence Queries (should NOT trigger HITL):

1. "How do I reset my password?"
   Expected: Detailed reset procedure

2. "What subscription plans are available?"
   Expected: List of Basic, Pro, Enterprise plans

3. "Is TechFlow GDPR compliant?"
   Expected: "Yes, TechFlow is fully GDPR compliant..."

4. "How do I enable push notifications?"
   Expected: Mobile app settings instructions

5. "What file formats are supported?"
   Expected: "PDF, DOCX, XLSX, CSV, PNG, JPG, TXT"

Low-Confidence Queries (should trigger HITL):

1. "What is the meaning of life?"
   Expected: "I don't have enough information..." + HITL escalation

2. "Can you recommend a good restaurant?"
   Expected: Out-of-scope + HITL escalation

3. "Help" (single word)
   Expected: Too vague, short answer + HITL escalation

4. "Why is my account locked?"
   Expected: Account-specific, requires human investigation + HITL

Edge Cases:

1. Empty query (user presses Enter)
   Expected: "Please enter a question."

2. Very long query (1000+ characters)
   Expected: Truncate to 500 chars, process

3. Special characters ("@#$%^&*()")
   Expected: Process normally (embedding handles it)

9. FUTURE ENHANCEMENTS

9.1 Multi-Document Support

Current Limitation: Single PDF only

Enhancement:

- Support multiple PDFs with metadata tagging
- Source attribution in answers ("According to Product_Manual.pdf, page 5...")
- Document prioritization (official docs > community wikis)

Implementation:

- Add "source" metadata to each chunk
- Filter retrieval by document type
- Display source citations in answers

9.2 Memory Integration

Current Limitation: No conversation history

Enhancement:

- Remember previous questions in the conversation
- Use context from prior exchanges
- Support follow-up questions

Implementation:

- Store conversation history in State
- Include previous Q&A pairs in retrieval context
- Use session management for multi-turn conversations

9.3 Feedback Loop

Current Limitation: No learning from user feedback

Enhancement:

- Collect user ratings on answers
- Use feedback to improve retrieval
- Identify knowledge gaps

Implementation:

- Add rating buttons to UI
- Store ratings in database
- Analyze low-rated answers to find patterns
- Retrain embedding model on feedback data

9.4 Deployment to Production

Current State: Local Python script

Enhancement:

- Deploy as web API (FastAPI/Flask)
- Containerize with Docker
- Host on cloud (AWS, GCP, Azure)
- Add authentication and rate limiting

Architecture:

- Frontend: React web app or mobile app
- Backend: FastAPI server
- Database: PostgreSQL for conversation logs
- Vector DB: ChromaDB or Pinecone
- Hosting: AWS EC2 or Lambda

10. REFERENCES

10.1 Research Papers

1. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" Lewis et al., 2020 https://arxiv.org/abs/2005.11401
2. "Dense Passage Retrieval for Open-Domain Question Answering" Karpukhin et al., 2020 https://arxiv.org/abs/2004.04906
3. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks" Reimers & Gurevych, 2019 https://arxiv.org/abs/1908.10084

10.2 Related Projects

- LlamaIndex: Alternative RAG framework
- Haystack: Open-source NLP framework with RAG support
- Weaviate: Vector database alternative to ChromaDB

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END OF HIGH-LEVEL DESIGN DOCUMENT